Part 2

Getting Started with GNU Octave

This time we'll explore matrices a little further and solve a few equations along the way.

In the first part of this series, we learnt how to work with matrices in Octave. We shall continue our journey with them and look at some more of their properties, such as condition numbers, norms and ranks. We will then solve a simple linear equation and touch upon matrix factorisation. I will try my best to keep things very simple. And when that's not possible, I will refer you to the appropriate references. Let's get started!

Common matrix properties

In this section I will discuss how to find some common matrix properties in Octave.

Norm: The norm of a matrix is a unique positive scalar number associated with it. Unlike the determinant, the norm is defined for all matrices—singular or non-singular, invertible or non-invertible. There are various ways to define the norm of a matrix. A very popular definition is the row sum norm. For an mXn matrix [A], the row sum norm of [A] is defined as:

$$\|A\| = \max_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|$$

That is, find the sum of the absolute values of the elements of each row, the maximum of these row sums is the *norm* of the matrix.

There are other possible definitions of the norm, like the Frobenius norm, which is very useful in numerical linear algebra and is defined as:

$$\|A\|_F = \sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2$$

In Octave, the command to find the norm of a matrix is *norm(A, p)*, where A is the matrix and p is the type of norm to be found. (p is optional and is 2, by default.)

If you want to find the row sum norm, *use p = Inf,* and for the Frobenius norm, use *p = "fro"*. For example, consider a matrix A:

```
octave:1> A= [ 1 , 2 ; 3 , 4 ]
A =

1  2
3  4
```

To find the *row sum norm* of A, use the following code:

```
octave:6> norm (A, I n f )
ans = 7
```

To find the *Fronbenius norm*, issue the following command:

```
octave:7> norm (A, ' ' f r o ' ' )
ans = 5.4772
```

To know the other possible values of p and what they do, please refer to the Octave user documentation.

Condition number: The physical significance of the *norm* depends on the type of *norm* we are considering. But at a higher